STILLSTACK · RISK REGISTER AND HAZARD ANALYSIS

StillStack risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: P7 and F11 with 5052 aluminium clips (D-08; F11 occurrence 3 to 1, RPN 48 to 16); P6 and F4 coupon test (D-04); H3 anchoring a hold point (D-09)
0.1	2026-10-24	Amish Chadha	First risk register, preliminary hazard analysis and design FMEA (kit 1.9.1 upgrade)

StillStack risk register and hazard analysis

The top risks are brine reaching the distillate (a health hazard if the water is drunk untested), a dry panel in full sun (burns and damage at about 142 °C), and a wick fabric that cannot carry the feed (output well below PR-001). Each has a control in the design and a pass condition or test planned for TRL 4. Method: STANDARDS section 23. Likelihood, impact, severity, occurrence and detection are scored 1 (low) to 5 (high; for detection, 5 means hardest to detect).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	Wick fabric too tight to carry the feed	Permeability below 3.0e-11 m2 (SSK-CAL-001 section 6)	Dry patches, output below PR-001, local stagnation	3	4	Design lead	Wicking-rise test of the two shortlisted fabrics before purchase (SSK-DEC-001, 2026-10-02)
P2	No certified food-contact coating found	Coaters may not certify hot salty distillate at 150 °C dry	PR-007 not met; plates 1 and 2 move to 316 stainless (cost up)	3	3	Design lead	Rule of 2026-10-02: lowest-cost certified coating, else 316 stainless for plates 1 and 2
P3	Cost stays over the value-engineering target	Coated plates and PPS rails	USD 105.70 over; harder for NGOs to fund	4	2	Design lead	Savings in SSK-DEC-001 (nested blanks, folded trim, reclaimed timber); report against the target
P4	Concept selection stays open	Options A and B within 10 % (SSK-OPT-001)	Effort spent on the wrong option	2	3	Amish	Decision D-03 requested
P5	No partner lab agrees to test	Partner not yet approached	TRL 4 delayed	3	3	Amish	First candidate named (Politecnico di Torino); a coastal NGO later
P7	Stainless clips corrode the plates	Galvanic couple of 304 and aluminium in hot brine (STEM S30)	Pitting at the clips; brine path to the distillate face	3	4	Design lead	5052 aluminium clips (D-08, decided by Amish 2026-10-25)
P6	Crevice corrosion perforates a plate	Bare aluminium under wet wicks and clips in 60 to 80 °C brine	Brine path to the distillate face; plate replacement	2	4	Design lead	Coupon test at TRL 4, 1050 preferred (D-04, decided by Amish 2026-10-25); 5052 as the fallback alloy

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Contaminated distillate drunk	Brine carryover, microbes below pasteurization, volatile organics in the feed	Illness	5	3	Separate distillate and brine tongues with 10 mm air breaks, lip tabs, labelled outlets (PR-004); test conductivity and E. coli and disinfect before drinking (PRC safety); per-stage conductivity pass condition at TRL 4	5 x 2: acceptable only with the testing rule in the instructions
H2	Hot surfaces	Absorber near 80 °C in use and 142 °C when dry (SSK-CAL-001 section 5)	Burns	3	3	Stagnation cover (BOM 14); never open the stack in sun; shade rule; trim and glazing kept on	3 x 2
H3	Panel tips over in wind	Unanchored, it tips at about 16.5 m/s of wind from behind; PR-013 is stated for the anchored stand (D-09)	Crush or cut injuries; broken glazing	4	3	Four ground stakes or 40 kg of ballast as a hold point before filling and before the panel faces the sun (BLD safety stops S4, S5; D-09, decided by Amish 2026-10-25); 28.8 m/s with the ballast	4 x 1 anchored
H4	Material damage at dry stagnation	Feed runs out on a sunny day	Softened rails, scorched wick, leaking stack	2	3	Ratings above stagnation in stages 1 to 3 (PPS since D-01), cover rule	2 x 2
H5	Brine discharge	Brine at 58 to 67 g/L	Salinised soil or wells	2	3	Discharge away from crops and wells (PRC safety; G10)	2 x 2
H6	Cuts during the build	0.5 mm aluminium sheet edges and tongues	Hand cuts	2	3	Deburr, gloves (BLD)	2 x 2
H7	Lifting the panel onto the stand	18.8 kg dry panel, awkward shape	Back strain, dropped panel	2	2	Two people, panel dry, cover on (BLD safety stop S4)	2 x 1

Design FMEA (first pass, P3)

Table 3. Failure modes. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Brine tongue and lid	Keep brine tails off the distillate path	Brine drips onto the lid slots	Salt in distillate (H1)	Tail sags, lid deformed	5	2	3	30	Lid slots only under distillate tongues; tails 10.5 mm clear; conductivity per stage at TRL 4
F2	Lip tabs on condensing plates	Stop feed creeping onto the condensing faces	Feed wets the condensing face at the high edge	Salt in distillate	Tabs bent wrong, wick tail touches plate	5	2	3	30	Bend before coating; check 1.5 mm clearance at assembly (model check)
F3	Wick strips	Carry feed down the slope	Dry patch	Lower output, local hot spot	Low permeability, salt crust	3	3	3	27	Fabric test; weekly rinse (PR-010)
F4	Plate (bare wick face)	Conduct heat, carry the wick	Crevice corrosion through 0.5 mm	Brine reaches distillate face	Hot brine crevice under wick and clips	5	2	4	40	Coupon test (D-04); inspect plates at each wick change
F5	Condensing-face coating	Food-contact surface	Coating blisters or leaches	PR-007 not met	Wrong product, poor adhesion on bends	4	2	3	24	Certified product only (rule of 2026-10-02); bend before coating
F6	Spacer ribs	Hold the 6 mm gap	Rib creeps, gap closes	Wick touches plate below; carryover	Silicone set under heat	3	2	3	18	Silicone rated about 200 °C; inspect gaps at wick change
F7	Side rails, stage 3	Hold the gap at the edges	Rail softens at stagnation	Gap closes at the edge	Printed PC at 110 °C (before D-01)	3	1	2	6	Closed by D-01 (PPS)
F8	Glazing (inner sheet)	Transmit sunlight, keep heat in	Sheet distorts	Lower output, leaks	115 °C at calm stagnation against a 120 °C limit	3	2	2	12	Cover rule (D-05)
F11	Wick clips	Hold the wicks	Clip or plate corrodes at the contact	Brine path or loose wick	Galvanic couple with stainless clips (removed by the 5052 aluminium clips, D-08, 2026-10-25);	4	1	4	16	5052 aluminium clips (D-08, decided by Amish 2026-10-25); coupon test (D-04); inspect

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
					crevice under the clip remains					at each wick change
F12	Fins and underside paint	Reject heat (emissivity 0.90, D-07)	Paint chalks or peels	Hotter stack, lower yield (about 1 L at emissivity 0.10)	UV and heat ageing	3	2	2	12	High-temperature paint as on the absorber; inspect yearly
F9	Stand foot pin	Hold the tilt	Pin missing or loose	Panel drops to the stops	Pin not refitted after a tilt change	4	2	2	16	M10 x 120 pins with nuts (D-02); two-person tilt change in BLD
F10	Ground stakes	Hold the stand in wind	Stake pulls out	Overturn (H3)	Soft or sandy ground	4	2	3	24	Ballast where stakes cannot hold; site check at TRL 5